

PEDRO PABLO ZAVALA TEJOS

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EDUCATION

Pontificia Universidad Católica de Chile 2021 – Jul. 2026
Santiago, Chile

Computer Science Engineer

- Major in Computing and Information Systems.
- Minor in Data Science and Analytics.
- Academic average: 6.22/7.00.
- Class rank: 47/950 – Top 5%.

PROFESSIONAL EXPERIENCE

Research Assistant Aug. 2025 – Present
Santiago / Bahía Exploradores, Chile

Estación Patagonia UC (EPII UC)

- PISMA-WP (Socio-environmental Monitoring Platform):** Design and develop data architecture and backend workflows for a socio-environmental monitoring platform, including AWS infrastructure using Terraform and Docker.
- ECHO/SAVIA (Remote Scientific Monitoring Infrastructure):** Maintain and extend an edge-to-cloud platform for remote scientific stations, improving data acquisition, synchronization, monitoring, and operational reliability.
- Wildlife Monitoring:** Contribute to a wildlife monitoring system using computer vision, human-in-the-loop workflows, and edge inference for remote environments.

Undergraduate Research Assistant Mar. 2025 – Jul. 2025
Santiago / Bahía Exploradores, Chile

Estación Patagonia UC (EPII UC), UC School of Engineering

- ECHO (Edge Computing and Hardware Orchestration):** Developed the initial platform for environmental data acquisition and monitoring in remote scientific stations, integrating heterogeneous data sources, APIs, sensors, and cloud storage using Python.

Research Assistant Aug. 2026 – Present
Santiago, Chile

IoT-UC Laboratory

- SCORPIO (Satellite LoRa Monitoring Platform):** Support deployment and evolution of a distributed edge-to-cloud platform for satellite LoRa measurements, maintaining installation tooling and containerized services and contributing to monitoring, debugging, and operation of distributed ground-station infrastructure.

Undergraduate Researcher Mar. 2026 – Jul. 2026
Santiago, Chile

IoT-UC Laboratory, UC School of Engineering

- Designed and implemented a modular edge-to-cloud architecture to acquire, process, back up, and centralize LoRa satellite data under intermittent connectivity.
- Developed Linux-based edge services in Python and Docker on Raspberry Pi using MQTT and SQLite, together with a REST API built with Node.js, TypeScript, Prisma, and PostgreSQL.

Degree Project – Natural-Language Data Query Assistant Mar. 2026 – Jun. 2026
Santiago, Chile

National Center for Artificial Intelligence (CENIA), UC School of Engineering

- Developed, in a six-member team, a conversational assistant that translates natural-language requests into validated SQL using Slack, RAG, PostgreSQL/pgvector, and automated AWS deployment with Terraform.

Data Engineering Intern Jan. 2025 – Apr. 2025
Santiago, Chile

SimplePark

- Developed and automated ETL pipelines and event-driven processes using Python, SQL, BigQuery, Cloud Functions, Pub/Sub, and Google Cloud Storage.
- Refactored data processes following Clean Architecture principles, incorporating validation and unit testing.

Capstone Project – AI-Assisted Academic Management Platform Aug. 2024 – Dec. 2024
Santiago, Chile

Faculty of Health Sciences, Pontificia Universidad Católica de Chile

- Implemented backend services using NestJS, TypeScript, Prisma, and PostgreSQL; integrated external sources and redesigned relational schemas for maintainability and scalability.

Teaching Assistant – Advanced Programming (IIC2233) Mar. 2023 – Jul. 2024
Santiago, Chile

Pontificia Universidad Católica de Chile

- Supported students during programming sessions and evaluated Python assignments and coursework.

PROFESSIONAL SKILLS

Programming & Automation: Python, TypeScript, SQL, scripting

Systems & Infrastructure: Linux, Docker, Terraform, GitHub Actions, CI/CD, Git/GitHub

Cloud & Processing: Google Cloud Platform, AWS, Cloud Functions, Pub/Sub

Backend & Integration: FastAPI, NestJS, REST APIs, service integration

Data & Storage: PostgreSQL, SQLite, BigQuery, MongoDB, Google Cloud Storage

Edge Data and IoT: MQTT/Mosquitto, Raspberry Pi, Home Assistant

HPC & Parallel Computing: MPI, OpenMP, Numba, SLURM

EXTRACURRICULAR ACTIVITIES

Speaker

ECHO: Monitoring and Automation Platform for Remote Stations

Ingeniería para Chile Conference, PUC

May 2026

LANGUAGES

Spanish: Native

English: B2 / Upper-intermediate proficiency

ADDITIONAL INFORMATION

Portfolio

pedrozavalat.github.io

LinkedIn

linkedin.com/in/pedro-pablo-zavala

GitHub

github.com/pedrozavalat

REFERENCES

Rodrigo Arnaldo Carrasco Schmidt

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